Asset ID: `DT-TRIPURA-001`

Dataset ID: `tripura_dev`

Report ID: `rep-DT-TRIPURA-001-000004`

State Timestamp: 2026-01-01 10:45:00 UTC

Watermark: 2026-01-01 10:35:00 UTC

DIGITAL TWIN AUDIT REPORT

POWERNEXT-AI HACKATHON 2026 · TRACK 2
ENGINEERING ASSET SUMMARY

1. EXECUTIVE HEALTH & OPERATING SUMMARY

OPERATIONAL HEALTH INDEX	EVIDENCE GRADE	TOP-OIL TEMPERATURE	LOADING & APPARENT POWER
0/100	**GRADE C**	**UNAVAILABLE**	**148.7 kVA**
CRITICAL	C1 Capability Tier	Expected: Unavailable (dT: N/A)	Load Index u: 1.10 p.u.

PRIMARY ACTION REQUIRED [P2 · NEXT_AUTHORIZED_MAINTENANCE_WINDOW]: Phase Imbalance / Neutral Path Maintenance — Inspect phase loading, connections, and neutral path during next authorized maintenance window.

Asset Profile & Lineage Context: This report represents an authoritative point-in-time digital twin state assessment for asset `DT-TRIPURA-001`. Telemetry is evaluated using a calibrated first-order thermal ODE (tau = 3.00 h) with locked 90% empirical conformal bounds and multi-domain OTHI penalty aggregation.

2. ENGINEERING STATE ASSESSMENT

Thermal Dynamics & Latent State

PARAMETER	OBSERVED / DERIVED VALUE
Top-Oil Observed Temperature	**Unavailable**
Top-Oil ODE Expected Temperature	Unavailable
Thermal Residual (T_obs - T_exp)	Unavailable
Thermal Time Constant (tau)	3.00 hours
Thermal Model Promotion Status	PROMOTED
90% Conformal Interval (Lower)	Unavailable
90% Conformal Interval (Upper)	Unavailable
Hot-Spot / WTI Temperature	**UNAVAILABLE (C4 GATED)**

Electrical Measurements & Features

SIGNAL / FEATURE	VALUE
Active Power (P Total)	Unavailable
Reactive Power (Q Total)	Unavailable
Apparent Power (S Derived)	148.7 kVA
Derived Power Factor (P/S)	0.895
Mean 3-Phase RMS Current	93.3 A
Current Magnitude Imbalance	50.00 % (NEMA)
Voltage Magnitude Imbalance	0.00 % (NEMA)
Neutral Conductor Current	Unavailable

PROVENANCE: State Hash: 2cf365af81de885b86811619d4b909afb9a20d89ea6fd234f602a0054e5f6609 | FactBundle Hash: 37a09f288e266e8d2c8ff0f0715967d8ba3543a9d25c158cd630fe4261aa12a9 | Seq: 4

MULTI-DOMAIN HEALTH BREAKDOWN

Asset ID: `DT-TRIPURA-001`

3. OPERATIONAL TRANSFORMER HEALTH INDEX (OTHI)

OTHI aggregates 5 canonical operational domains via a non-linear gated max-plus-secondary penalty formula: *Penalty = min(1.0, max(s_d) + 0.20 * sum(other s_d))*.

DOMAIN	STATUS	SEVERITY (S_D)	FRESHNESS	CONTRIBUTING FAMILIES	ENGINEERING EXPLANATION
thermal	NORMAL	0.00	CURRENT	None	thermal condition: severity 0.00 (NORMAL), families: ['NORMAL']
loading	WARNING	0.60	CURRENT	LOADING	loading condition: severity 0.60 (WARNING), families: ['LOADING']
phase_neutral	CRITICAL	0.90	CURRENT	PHASE_CURRENT_IMBALANCE	phase_neutral condition: severity 0.90 (CRITICAL), families: ['PHASE_CURRENT_IMBALANCE']
voltage_pf_freq	NORMAL	0.00	CURRENT	None	voltage_pf_freq condition: severity 0.00 (NORMAL), families: ['NORMAL']
oil_protection	NORMAL	0.00	UNAVAILABLE	None	oil_protection condition: severity 0.00 (NORMAL), families: ['NORMAL']

Health Gating & Anti-Uplift Guarantee

Is Gated: NO (Score Active)
Gating Reason: Multi-domain evidence threshold satisfied.
Driving Domain: phase_neutral
Base Penalty: 0.900
Secondary Penalty: 0.120
Total Penalty: 1.000

Statistical Detectors & CUSUM

CUSUM Pos Accumulator: 0.000
CUSUM Neg Accumulator: 0.000
Decision Threshold Tripped: CLEAR
Active Persistence Timers: 1 active counters

4. ACTIVE ANOMALY INCIDENTS (6-STATE FSM)

EPISODE ID	CATEGORY	STATE	SEVERITY	START (UTC)	PRIMARY REASON & HYPOTHESES
ep_DT-TRIPURA-001_PHASE_IMBALANCE_2[illegible]04500_1	PHASE_IMBALANCE	WATCH	0.90	2026-01-01 10:45	**CRITICAL: Three-phase current magnitude unbalance reached 50.0% (NEMA limit exceeded).** Hypotheses: UNBALANCED_SINGLE_PHASE_L LOOSE_TERMINAL_CONNECTION ASYMMETRIC_FEEDER_DEMAND

5. PRIORITIZED MAINTENANCE ACTION PLAN

PRIORITY	WINDOW	ACTION TITLE	ADVISORY WORDING & RATIONALE	STANDARD REF
P2	NEXT_AUTHORIZED_MAINTENANCE_WINDOW	**Phase Imbalance / Neutral Path Maintenance** ID: REC-DT-TRIPURA-001-MNT-PHASE-004-1767284300 Rule: MNT-PHASE-004	Inspect phase loading, connections, and neutral path during next authorized maintenance window. *Rationale: Persistent current imbalance corroborated by neutral return creates uneven winding thermal stress and stray losses.*	Feeder Phase Balancing Maintenance Standard
P4	MODEL_REVIEW	**Sensor Plausibility / Flatline Fault** ID: REC-DT-TRIPURA-001-MNT-SENS-005 Rule: MNT-SENS-006	Inspect identified sensor and mapping. Model-dependent conclusions suspended until data quality recovers. *Rationale: Sensor flatline, step anomaly, or unphysical value detected. Suspends model dependency until channel is verified.*	Telemetry Data Governance & Sensor Validation Standard

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HISTORICAL TRENDS & AUDIT ANNEX

Asset ID: DT-TRIPURA-001

6. AUTHORITATIVE HISTORICAL TRENDS

Historical telemetry frames unavailable for chart rendering in current storage context.

7. ENGINEERING ANNEX & STANDARDS COMPLIANCE

Standards & Citations

• **IEEE C57.91-2011:** Guide for Loading Mineral-Oil-Immersed Transformers (Thermal ODE reference & insulation aging models).
• **IEC 60076-7:** Loading guide for oil-immersed power transformers.
• **NEMA MG-1:** Phase voltage and current unbalance calculation conventions.

Safety & Data Honesty Guarantees

• **Zero Unsupported Claims:** No numerical Remaining Useful Life (RUL) or autonomous trip commands.
• **Unverified Nameplate Rating:** Relative load index (u) is reported in per-unit reference without fabricated kVA capacity.
• **Model Lineage:** Thermal ODE is qualified under descriptive shadow monitoring.

PROVENANCE AUDIT: State Hash: 2cf365af81de885b86811619d4b909afb9a20d89ea6fd234f602a0054e5f6609 | FactBundle Hash: 37a09f288e266e8d2c8ff0f0715967d8ba3543a9d25c158cd630fe4261aa12a9 | Generated At: 2026-01-01 10:45:00 UTC